

PCA (Principal Component Analysis) Complete Analysis Guide

A Step-by-Step Geometric and Mathematical Breakdown using a 5-Point Dataset

1. The 5-Point Student Score Dataset

We begin with a 2D dataset representing 5 students' Math (X) and Physics (Y) scores. The goal of PCA is to find the main axes of variance to reduce dimensionality and filter noise.

Student	Math Score (X)	Physics Score (Y)
A	2.0	3.0
B	3.0	4.0
C	5.0	5.0
D	6.0	7.0
E	9.0	6.0

2. Mathematical Decomposition Steps

Step 1: Mean Centering

Calculate mean values (Mean X = 5.0, Mean Y = 5.0) and subtract them from all points to shift the origin to (0,0).
Centered coordinates: A(-3, -2), B(-2, -1), C(0, 0), D(1, 2), E(4, 1).

Step 2: Covariance Matrix Generation

Measure how variables disperse and correlate. The computed Sample Covariance Matrix (Σ) is:
 $\Sigma = \begin{bmatrix} 7.5 & 3.5 \\ 3.5 & 2.5 \end{bmatrix}$

Step 3: Eigen-Decomposition

Solve $\det(\Sigma - \lambda I) = 0$ to extract eigenvalues (λ) and eigenvectors (v).

• **First Principal Component (PC1):** $\lambda_1 = 9.38$ (Explains **93.8%** of total variance). Direction Vector $v_1 = [0.882, 0.471]^T$.

• **Second Principal Component (PC2):** $\lambda_2 = 0.62$ (Explains **6.2%** of total variance). Direction Vector $v_2 = [-0.471, 0.882]^T$.

3. The 'Shape Magnifier' Concept & Linear Transformation

Multiplying data vectors by the covariance matrix acts as a geometric transformation that stretches the data along its structural skeleton.

If we feed the centered coordinates into Σ ($\vec{x}' = \Sigma \vec{x}$), we witness a massive unscaled directional pull towards the PC1 axis line (slope ≈ 0.53):

- Original Centered E(4, 1) $\rightarrow$ **True Transformed E'(33.5, 16.5)**
- Original Centered A(-3, -2) $\rightarrow$ **True Transformed A'(-29.5, -15.5)**

The covariance matrix amplifies the core underlying trend by $\lambda_1 = 9.38$, turning minor variations into an aligned, prominent straight-line projection.

4. Data Denoising & Optimal Reconstruction

By discarding PC2 ($\lambda_2 = 0.62$) as random testing error or score noise, we can clean and reconstruct the original data points using only PC1 attributes through Low-Rank Approximation:

Formula: Reconstruction = (Centered_x $\cdot$ $\mathbf{v}_1$) $\times$ $\mathbf{v}_1$ + Mean

- **Student A Reconstruction:** $(-3.59) \times [0.882, 0.471]^T + [5, 5]^T = \mathbf{(1.83, 3.31)}$ (Original: 2, 3)
- **Student E Reconstruction:** $(4.00) \times [0.882, 0.471]^T + [5, 5]^T = \mathbf{(8.53, 6.88)}$ (Original: 9, 6)

Insight: Random individual anomalies (e.g., Student E's outstanding math score compared to physics) are filtered out, pulling all coordinates perfectly back onto the main 'Science Aptitude Line' while retaining key individual variance.

This document compiles the conceptual framework, structural matrices, and mathematical denoising models discussed in the PCA query sessions.